

Assignment : Day 2

Diagnosing Hardware Failures
Theory : Common hardware issues and their symptoms.
Practical : Test components using a multimeter.

TASK : 1

List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).



Hardware Issue -> Symptoms

No.	Hardware Issues	Symptoms
1	RAM Failure	No Display, Repeated Beep Sound
2	HDD/SSD Failure	Fail to boot ,PC may become very slow
3	CPU Overheating	Display freezes , work unstable
4	GPU Failure	Distorted Graphics , No display
5	SMPS Failure	PC may not start or suddenly shut down

If something fails , symptoms may guide you to check which component is bringing problem

TASK : 2

Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.



The goal is to find out whether the problem is with the **power source, charger, charging port, battery, or internal connection**.

Step-by-step troubleshooting plan

Step 1: Check the wall socket

- Plug another working device into the same socket.

- If the other device also doesn't work, the problem may be with the socket or power supply.
- Try another known-working socket.

Step 2: Inspect the charger

- Check the charger cable for cuts, bends, burns, or exposed wires.
- Check the charging adapter for physical damage or unusual smell.
- Make sure the cable is firmly connected to the adapter.

Step 3: Check the laptop charging port

- Look inside the charging port for dust, dirt, bent pins, or physical damage.
- Make sure the charger connector fits properly and isn't unusually loose.

Step 4: Check the charger output with a multimeter

- Set the multimeter to **DC voltage (V=)**.
- Check the charger's output using its connector, following the charger's polarity markings.
- Compare the measured voltage with the **output voltage printed on the charger label**.
- A significantly incorrect or zero reading can indicate a faulty charger.

Step 5: Check the battery

- If the laptop has a removable battery, inspect it for physical damage or swelling.
- If the battery is swollen, **do not use or puncture it**; it should be handled/replaced safely.
- If possible, test the laptop with a known-good compatible battery or charger.

Step 6: Check the connection

- If the charger and battery appear normal, the problem could be an internal charging circuit, DC jack, connector, or motherboard component.
- These internal components should be checked by a qualified technician rather than probing randomly.

Simple troubleshooting flow

Wall socket → Charger → Charging port → Charger voltage → Battery → Internal charging circuit

TASK : 3

Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



A standard USB charger normally provides approximately:

5V DC so expected range is:

4.75V to 5.25V

So:

- **4.75–5.25V → Within the expected range**
- **Below 4.75V → Lower than expected**
- **Above 5.25V → Higher than expected**

Set the multimeter to DC Voltage , than connect the Black Probe to COM and Red Probe to V input and put the adapter to the power outlet

For a standard USB-A output:

- **Red probe → +5V/VBUS**
- **Black probe → GND**

A multimeter can be used to check the DC voltage output of a USB phone charger. A standard USB output is expected to be approximately 5V DC. The measured value can be compared with the expected range of 4.75V to 5.25V to determine whether the charger is providing a normal output

TASK : 4

A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.



This task is about SMPS/ATX PSU is supplying a correct 12 v power supply because if not sytem may freeze or shut down

In standard ATX power supply:

Yellow wire = 12V

Black wire = Ground

The expected 12V Rail is under 11.4 to 12.6 under the ATX tolerance +5%

Step-by-step procedure

Step 1: Turn off the PC

- Shut down the computer completely.
- Switch off and unplug the PSU from the wall.

Step 2: Disconnect the ATX connector

- Disconnect the ATX power connector from the motherboard.
- Do not perform this measurement while the connector is connected to the motherboard unless following a proper PSU-testing procedure.

Step 3: Set the multimeter

- Set the multimeter to **DC voltage (V=)**.
- Connect the **black probe to COM**.
- Connect the **red probe to V**.

Step 4: Identify the wires

- Find a **yellow wire** in the ATX connector.
- Find a **black wire**, which is ground.

Step 5: Measure the voltage

- Place the **red probe on the yellow (+12V) wire/pin**.
- Place the **black probe on the black (GND) wire/pin**.
- Record the voltage shown on the multimeter.

Compare the Reading

Measured Voltage	Interpretation
Around 12V	Normal 12V rail
11.5 TO 11.9	Low but above assignment threshold
Lower than 11.5	Weak power supply faulty unstable

